

# Lening Li

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## RESEARCH INTERESTS

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Reinforcement Learning ◊ Optimal Control ◊ Game Theory ◊ Formal Methods

## EDUCATION

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**Carnegie Mellon University (CMU)**

Pittsburgh, PA, USA

**MBA**

Aug. 2025 - Present (*Expected May 2028*)

**Worcester Polytechnic Institute (WPI)**

Worcester, MA, USA

**Ph.D. in Robotics Engineering**

Aug. 2016 - Dec. 2022

- ◊ Thesis: “Optimal Control and Reinforcement Learning for Stochastic Systems under Temporal Logic Specifications”

**M.S. in Robotics Engineering**

Aug. 2016 - May 2018

**M.S. in Computer Science**

Aug. 2014 - May 2016

- ◊ Thesis: “Birrtopt: A combined software framework for motion planning applied on Atlas robot”

**Harbin Institute of Technology (HIT)**

China

**B.S. in Computer Science**

Sep. 2010 - Jul. 2014

- ◊ Summa Cum Laude (Top 5% of class)
- ◊ Thesis: “Contourlet Transform Based Image Compression”

**B.A. in English Language & Literature**

Sep. 2011 - Jul. 2014

- ◊ Thesis: “A Study on the Male Chauvinism in *Women in Love*”

## CERTIFICATIONS

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**Certified REC Foundation Coach**

Sep. 2023 - Sep. 2024

*Robotics Education & Competition Foundation*

Online

Coach and mentor robotics teams, fostering student growth in STEM and competitive engineering.

**Certified FIRST Tech Challenge (FTC) Coach**

Nov. 2022 - May 2023

*FIRST Robotics*

Online

Led and mentored FTC robotics teams, guiding students in robot design, programming, and competition strategy.

**Certification in College Teaching**

Jun. 2017 - Aug. 2019

*Higher Education Consortium of Central Massachusetts (HECCMA)*

Worcester, MA, USA

Professionally trained in evidence-based pedagogy to deliver high-quality courses.

## INDUSTRY & ADVISORY EXPERIENCE

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**Robotics Lab Advisor**

Dec. 2025 - Present

*Harvard University*

Cambridge, MA, USA

- ◊ Guide student researchers on algorithm design, system evaluation, and reproducible methodology for robotics projects.
- ◊ Mentor students across reinforcement learning, motion planning, control, and safety-critical system design, connecting theory to production deployment.

- Accelerate research output through structured reviews, project scoping, and targeted technical feedback.

## Senior Software Engineer

*Symbolic*

Oct. 2022 - Present  
Wilmington, MA, USA

- Designed and deployed production multi-agent path-planning and coordination algorithms in C++ for warehouse robot fleets of thousands.
- Improved fleet robustness through advances in control design, state estimation, and fault-tolerant behavior architectures.
- Optimized real-time decision-making pipelines to meet strict latency requirements across thousands of concurrent autonomous robots.

## Senior Software Engineer

*Berkshire Grey*

Oct. 2021 - Aug. 2022  
Bedford, MA, USA

- Led development of perception and manipulation algorithms enabling robust grasping of novel, previously unseen SKUs (ROS, C++, Python).
- Cut end-to-end system latency by overhauling inter-process communication and execution pipelines.

## Software Engineering Intern

*Rudolph Technologies*

Jun. 2015 - Jan. 2016  
Tewksbury, MA, USA

- Built an automated migration tool consolidating multiple legacy codebases onto a single unified platform.
- Improved wafer-defect data accuracy through algorithmic enhancements to collection and analysis pipelines.

## Software Engineering Intern

*Neusoft*

Jul. 2013 - Aug. 2013  
China

- Developed a map management system with efficient insert, delete, and update operations on geographic datasets.

## REPRESENTATIVE SKILLS

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|--------------------------------|----------------------------------------------------------|
| <b>Programming Languages:</b>  | C/C++, Python, MATLAB                                    |
| <b>Robotics &amp; Control:</b> | Motion Planning, Optimal Control, Reinforcement Learning |
| <b>Robotic Systems:</b>        | ROS, ROS 2, Distributed Robotic Systems                  |
| <b>Machine Learning:</b>       | PyTorch, TensorFlow                                      |
| <b>Languages:</b>              | English (Fluent), Chinese (Native)                       |

## TEACHING

### Teaching Assistant

*RBE 549. Computer Vision, WPI*

Aug. 2022 - Dec. 2022  
Worcester, MA, USA

- Co-designed and delivered a new graduate-level computer vision course, covering classical and deep learning approaches.
- Led weekly lectures and structured office hours, supporting student project development.
- Course website: <https://nitinjsanket.github.io/teaching/rbe549/fall2022.html>

### Teaching Assistant

*RBE 3001 & 3002. Unified Robotics III & IV, WPI*

Aug. 2020 - May 2021  
Worcester, MA, USA

- Supervised hands-on labs covering 3D-printed robot-arm kinematics, control, and mobile robot navigation.

- Designed and graded final projects, lab reports, and homework assignments.

### **Teaching Assistant**

*RBE 549. Computer Vision, WPI*

Aug. 2018 - Dec. 2018

Worcester, MA, USA

- Delivered lectures and held office hours on computer vision theory and implementation for graduate students.

### **Teaching Assistant**

*RBE 1001. Introduction to Robotics, WPI*

Aug. 2017 - May 2018

Worcester, MA, USA

- Managed and mentored a team of five undergraduate peer learning assistants.
- Designed and graded final projects, lab reports, and homework assignments.

## **REPRESENTATIVE PROJECTS**

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### **DARPA Robotics Challenge**

*Researcher*

Aug. 2014 - May 2015

Worcester, MA, USA

- Collaborated with the WPI-CMU team on Boston Dynamics' Atlas humanoid robot at the DARPA Robotics Challenge Finals.
- Designed a motion planner for arm manipulation tasks including door opening, valve turning, and tool grasping.
- Designed a human-robot interaction interface that improved operator situational awareness and task execution efficiency.
- Ranked 7th out of 24 teams.

### **DARPA SI3-CMD: Serial Interactions in Imperfect Information Games for Complex Military Decision Making**

*Researcher*

Jan. 2019 - Aug. 2020

Worcester, MA, USA

- Developed a game-theoretic deceptive planning framework and Python implementation in collaboration with SSCI.
- Improved mission success probability by exploiting information asymmetry and strategic deception within a formal hypergame model.
- Formulated a solution concept for dynamic hypergames with temporal objectives.

### **Optimal Control and Reinforcement Learning for Stochastic Systems under Temporal Logic Specifications**

*Researcher*

Aug. 2016 - Aug. 2022

Worcester, MA, USA

- Developed a principled framework translating probabilistic temporal logic specifications into chance-constrained control problems with satisfaction guarantees.
- Proposed a scalable, model-free reinforcement learning approach for continuous stochastic systems with improved sample efficiency.

## **HONORS & AWARDS**

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### **Alex F. Backlin Fund Scholarship**

*Worcester Polytechnic Institute*

Jan. 2021

Worcester, MA, USA

### **Travel Grant Award**

*Lehigh University*

Oct. 2019

Bethlehem, PA, USA

### **Graduate Student Travel Award**

*Worcester Polytechnic Institute*

Oct. 2019

Worcester, MA, USA

**Graduate Student Travel Award**  
*Worcester Polytechnic Institute*

Mar. 2019  
Worcester, MA, USA

**Graduate Student Travel Award**  
*Worcester Polytechnic Institute*

Jun. 2017  
Worcester, MA, USA

## EXTRACURRICULAR ACTIVITIES AND LEADERSHIP

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**President**  
*Graduate Student Government (GSG)*

Jan. 2019 - May 2020  
Worcester, MA, USA

- Led WPI's Graduate Student Government, overseeing all governance operations and institutional representation.
- Served as primary liaison between the graduate student body and university administration.
- Partnered with the Graduate Studies Office to escalate housing concerns directly to the Board of Trustees.
- Appointed to the Provost Search Committee and provided formal recommendations.

**Volunteer**  
*Lhasa Welfare Center for Children*

Jul. 2013 - Sep. 2013  
China

- Fundraised to support educational services for children from low-income families.
- Tutored children in Chinese, Math, and English.

## ORGANIZATION MEMBERSHIPS

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- **Member**, IEEE
- **Member**, IEEE Young Professionals
- **Member**, IEEE Robotics and Automation Society
- **Member**, Association for Women in Mathematics
- **Member**, Alpha Chapter of Rho Beta Epsilon (Honor society) at Worcester Polytechnic Institute

## ACADEMIC SERVICES

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### Journal Reviewer

- IEEE Robotics and Automation Letters (RA-L)
- IET Cyber-Systems and Robotics
- IEEE Transactions on Intelligent Transportation Systems
- Discover Robotics
- The Journal of Supercomputing

### Conference Reviewer

- International Conference on Robotics and Automation (ICRA)
- American Control Conference (ACC)
- IEEE Conference on Decision and Control (CDC)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Conference on Ubiquitous Robots (UR)
- European Control Conference (ECC)

## SOCIAL MEDIA

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**LinkedIn:** <https://www.linkedin.com/in/lening-li/>  
**Profile:** <https://lening.li>  
**GitHub:** <https://github.com/leelening>  
**Google Scholar:** <https://scholar.google.com/citations?user=KWUJ10wAAAAJ\&hl=en>

## PUBLICATIONS

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### arXiv Preprints

- P.1 **L. Li** and Z. Qian, “Topological guided actor-critic modular learning of continuous systems with temporal objectives,” *arXiv preprint arXiv:2304.10041*, 2023
- P.2 **L. Li** and J. Fu, “Policy synthesis for metric interval temporal logic with probabilistic distributions,” *arXiv preprint arXiv:2105.04593*, 2021

### Conferences

- C.1 C. G. Atkeson, B. P. W. Babu, N. Banerjee, D. Berenson, C. P. Bove, X. Cui, M. DeDonato, R. Du, **L. Li**, P. Franklin, *et al.*, “No falls, no resets: Reliable humanoid behavior in the darpa robotics challenge,” in *2015 IEEE-RAS 15th International Conference on Humanoid Robots (Humanoids)*, pp. 623–630, IEEE, 2015
- C.2 **L. Li**, X. Long, and M. A. Gennert, “Birrtopt: A combined sampling and optimizing motion planner for humanoid robots,” in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, pp. 469–476, IEEE, 2016
- C.3 **L. Li** and J. Fu, “Sampling-based approximate optimal temporal logic planning,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 1328–1335, IEEE, 2017
- C.4 **L. Li** and J. Fu, “Topological approximate dynamic programming under temporal logic constraints,” in *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 5330–5337, IEEE, 2019
- C.5 **L. Li** and J. Fu, “Approximate dynamic programming with probabilistic temporal logic constraints,” in *2019 American Control Conference (ACC)*, pp. 1696–1703, IEEE, 2019
- C.6 **L. Li**, H. Ma, S. Han, and J. Fu, “Synthesis of proactive sensor placement in probabilistic attack graphs,” in *2023 American Control Conference (ACC)*, pp. 3415–3421, IEEE, 2023
- C.7 **L. Li**, H. Rahmani, and J. Fu, “Probabilistic planning with prioritized preferences over temporal logic objectives,” in *32nd International Joint Conference on Artificial Intelligence*, 2023(Acceptance rate: **15%**)

### Journals

- J.1 M. DeDonato, F. Polido, K. Knoedler, B. P. Babu, N. Banerjee, C. P. Bove, **L. Li**, R. Du, P. Franklin, J. P. Graff, *et al.*, “Team wpi-cmu: Achieving reliable humanoid behavior in the darpa robotics challenge,” *Journal of Field Robotics*, vol. 34, no. 2, pp. 381–399, 2017
- J.2 Z. Chen, **L. Li**, and X. Huang, “Building an autonomous lane keeping simulator using real-world data and end-to-end learning,” *IEEE Intelligent Transportation Systems Magazine*, vol. 12, no. 1, pp. 47–59, 2018
- J.3 **L. Li**, H. Ma, A. N. Kulkarni, and J. Fu, “Dynamic hypergames for synthesis of deceptive strategies with temporal logic objectives,” *IEEE Transactions on Automation Science and Engineering*, 2022

### Chapters

- Ch.1 C. G. Atkeson, P. B. Benezun, N. Banerjee, D. Berenson, C. P. Bove, X. Cui, M. DeDonato, R. Du, **L. Li**, P. Franklin, *et al.*, “Achieving reliable humanoid robot operations in the darpa robotics challenge: Team wpi-cmu’s approach,” in *The DARPA Robotics Challenge Finals: Humanoid Robots To The Rescue*, pp. 271–307, Springer, 2018

Ch.2 C. G. Atkeson, P. B. Benzun, N. Banerjee, D. Berenson, C. P. Bove, X. Cui, M. DeDonato, R. Du, **L. Li**, P. Franklin, *et al.*, “What happened at the darpa robotics challenge finals,” in *The DARPA Robotics Challenge Finals: Humanoid Robots To The Rescue*, pp. 667–684, Springer, 2018